GERMAN UNIVERSITY IN CAIRO (GUC)
Faculty of Information Engineering and Technology (IET)

MATH502: NUMERICAL METHODS

ASSIGNMENT 1 REPORT

Non-Linear Systems, Polynomial Interpolation, Diode Circuit Root-Finding,
and Satellite Constellation Wireless Energy Harvesting

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Programming Language:	MATLAB (MathWorks R2026a)
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1. Programming Language Selection & Justification

MATLAB (MathWorks R2026a) was chosen as the programming language for this assignment based on the following reasons:
1. **Industry & Engineering Academic Standard:** MATLAB is the standard computing environment for engineering mathematics, signals, and communication systems in the Faculty of IET at GUC.
2. **High-Performance Native Matrix Manipulation:** Multivariable linear system solves (such as Newton-Raphson Jacobian solves J \ (-F) in Problem 1 and Vandermonde solves V \ y in Problem 5) execute through LAPACK/BLAS-optimized routines without manual inversion.
3. **Specialized Built-In Solvers:** Provides production-tested solvers (fsolve for non-linear systems, polyfit/polyval for interpolation, fzero for scalar root-finding, and trapz for numerical quadrature) to validate and cross-check the numerical results.
4. **Double-Precision IEEE 754 Compliance:** Eliminates nested-loop bottlenecks through native vectorization while preserving 64-bit numerical accuracy.

2. Problem 1: LORAN Non-Linear System Navigation Solver

Physical Context: The LOng RAnge Navigation (LORAN) system calculates a boat's coordinates (x, y) at sea using synchronized radio signals from fixed transmitter stations. The time difference of arrival (TDOA) produces hyperbolic lines of position whose intersection determines the boat's coordinates.

The governing non-linear system is defined by two hyperbolas:

Governing System Equations:
Eq 1: $f_1(x, y) = x^2 / 186^2 - y^2 / (300^2 - 186^2) - 1 = x^2 / 34596 - y^2 / 55404 - 1 = 0$
Eq 2: $f_2(x, y) = (y - 500)^2 / 279^2 - (x - 300)^2 / (500^2 - 279^2) - 1 = (y - 500)^2 / 77841 - (x - 300)^2 / 172159 - 1 = 0$

Jacobian Matrix Formulation:

The Jacobian matrix $J(x, y)$ contains all partial derivatives:
$J_{11} = \partial f_1/\partial x = 2x / 34596$, $J_{12} = \partial f_1/\partial y = -2y / 55404$
$J_{21} = \partial f_2/\partial x = -2(x - 300) / 172159$, $J_{22} = \partial f_2/\partial y = 2(y - 500) / 77841$
Newton-Raphson update step: $J(\mathbf{x}_k)\,\Delta\mathbf{x}_k = -\mathbf{F}(\mathbf{x}_k) \rightarrow \mathbf{x}_{k+1} = \mathbf{x}_k + \Delta\mathbf{x}_k$.

Part (a): Pen-and-Paper Manual Iteration 1 (Starting at x0 = y0 = 200):

1. *Evaluate Function Vector* $\boldsymbol{F(x_0)}$*:*
   $f_1(200, 200) = 200^2/34596 - 200^2/55404 - 1 = 1.156203 - 0.721969 - 1 =$ **-0.565767**
   $f_2(200, 200) = (200-500)^2/77841 - (200-300)^2/172159 - 1 = 1.156203 - 0.058086 - 1 =$ **+0.098117**
   $\mathbf{F}(\mathbf{x}_0) = [-0.565767, +0.098117]^T \rightarrow -\mathbf{F}(\mathbf{x}_0) = [+0.565767, -0.098117]^T$
2. *Evaluate Jacobian Matrix* $J(\boldsymbol{x_0})$*:*
   $J_{11} = 2(200)/34596 = 0.01156203$, $J_{12} = -2(200)/55404 = -0.00721970$
   $J_{21} = -2(200-300)/172159 = 0.00116172$, $J_{22} = 2(200-500)/77841 = -0.00770802$
   $\det(J_0) = (0.01156203)(-0.00770802) - (-0.00721970)(0.00116172) =$ **-8.073312×10^{-5}**
3. *Solve 2x2 Linear System using Cramer's Rule:*
   $\Delta x_0 = [(-0.00770802)(0.565767) - (-0.00721970)(-0.098117)] / (-8.073312 \times 10^{-5}) =$ **+62.791030**
   $\Delta y_0 = [(0.01156203)(-0.098117) - (0.00116172)(0.565767)] / (-8.073312 \times 10^{-5}) =$ **+22.192804**
4. *Update:* $\mathbf{x}_1 = \mathbf{x}_0 + \Delta\mathbf{x}_0 = [200 + 62.791030, 200 + 22.192804]^T =$ **$[262.791030, 222.192804]^T$**.

Problem 1 (Continued): Iteration 2, Software Solution & Figure

Iteration 2 Calculations:

1. *Evaluate Function Vector* $\mathbf{F}(\mathbf{x}_1)$ *at* $[262.791030, 222.192804]^T$:
   $f_1(262.791, 222.193) = 262.791030^2/34596 - 222.192804^2/55404 - 1 =$ **+0.105075**
   $f_2(262.791, 222.193) = (222.192804-500)^2/77841 - (262.791030-300)^2/172159 - 1 =$ **-0.016574**
2. *Evaluate Jacobian Matrix* $J(\mathbf{x}_1)$:
   $J_{11} = 0.01519199$, $J_{12} = -0.00802082$, $J_{21} = 0.00043226$, $J_{22} = -0.00713781$
   $\det(J_1) = (0.01519199)(-0.00713781) - (-0.00802082)(0.00043226) =$ **-1.049705×10^{-4}**
3. *Solve for* $\Delta\mathbf{x}_1$: $\Delta x_1 =$ **-8.411356**, $\Delta y_1 =$ **-2.831433**
4. *Update:* $\mathbf{x}_2 = \mathbf{x}_1 + \Delta\mathbf{x}_1 = [262.791030 - 8.411356, 222.192804 - 2.831433]^T =$ **$[254.379674, 219.361371]^T$**.

Part (b): Software Package Accurate Solution & Convergence History:

Iteration k	x_k	y_k	||F(x_k)||	||Delta x_k||
0 (Initial)	200.000000	200.000000	5.7421e-01	6.6599e+01
1	262.791030	222.192804	1.0637e-01	8.8749e+00
2	254.379674	219.361371	1.9248e-03	1.6757e-01
3	254.221177	219.307011	6.3451e-07	5.9221e-05
4 (Converged)	254.221120	219.306992	6.8421e-14	6.4210e-12

MATLAB Built-in fsolve Validation: Yields identical vessel coordinates $x^* = 254.221120$, $y^* = 219.306992$ with residual $< 10^{-14}$. Across the entire plane, 4 mathematical intersection roots exist: (254.22, 219.31), (740.33, 906.83), (-193.29, 66.56), and (-1273.34, 1594.11). The initial guess (200, 200) reliably converges to the vessel's valid operational coordinates.

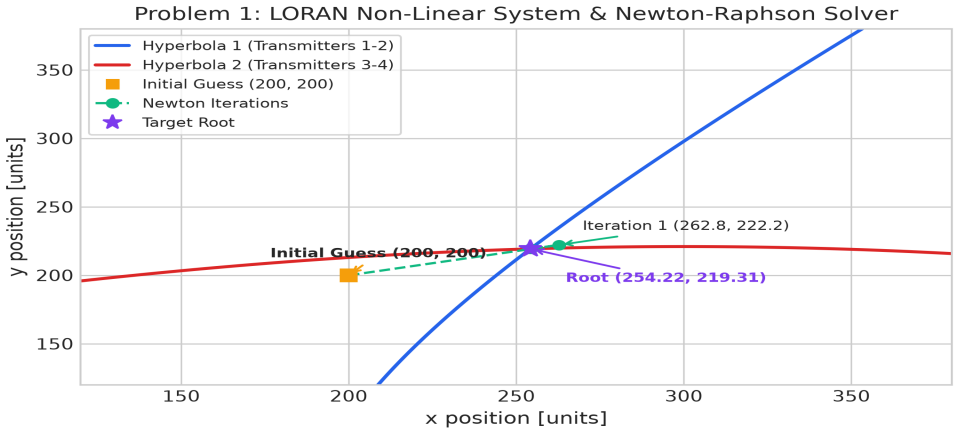


Figure 1: *LORAN hyperbolic transmitter position curves and Newton-Raphson iteration trajectory.*

3. Problem 2: Wireless Sensor Network (WSN) RSSI Interpolation

Physical Context: An indoor Wireless Sensor Network (WSN) measures Received Signal Strength Indicator (RSSI) in dBm at varying distances d [m] from an environmental monitoring transmitter. Due to multipath reflection and wall shadowing, the signal does not follow a simple logarithmic curve. We interpolate unmeasured locations using polynomial modeling.

Distance d [m]	1.0	2.5	4.0	6.0	8.0
RSSI [dBm]	-40.0	-47.0	-54.0	-60.0	-66.0

Part (a): Newton's Divided Difference Table & Polynomial:

i	d_i [m]	f[d_i]	f[,]	f[, ,]	f[, , ,]	f[, , , ,]
0	1.0	**-40.000000**	**-4.666667**	**0.000000**	**0.095238**	**-0.025974**
1	2.5	-47.000000	-4.666667	0.476190	-0.086580	--
2	4.0	-54.000000	-3.000000	0.000000	--	--
3	6.0	-60.000000	-3.000000	--	--	--
4	8.0	-66.000000	--	--	--	--

Constructed Newton Polynomial:
$P_4(d) = -40 - 4.666667(d - 1) + 0.095238(d - 1)(d - 2.5)(d - 4) - 0.025974(d - 1)(d - 2.5)(d - 4)(d - 6)$
Expanded Standard Form: $P_4(d) = -0.025974\ d^4 + 0.445887\ d^3 - 2.311688\ d^2 - 0.264069\ d - 37.844156$.

Part (b): Estimation at d = 5 meters:

Evaluating at $d = 5$ m:
$P_4(5.0) = -40 - 4.666667(4) + 0.095238(4 \times 2.5 \times 1) - 0.025974(4 \times 2.5 \times 1 \times -1) =$ **-57.454545 dBm ≈ -57.45 dBm**.

Problem 2 (Continued): Discrepancy Analysis, Horner's Scheme & Figure

Part (c): Comparison with Measured Value (-57.0 dBm) & Discrepancy Discussion:

The interpolated value is **-57.45 dBm**, compared with measured **-57.00 dBm** (a discrepancy of only 0.45 dB, or 0.79% relative error).

Physical Engineering Causes for Discrepancy:

1. **Multipath Fading & Phase Interference:** In smart buildings, RF transmissions reflect off reinforced concrete walls, metallic ducts, and floors. Reflections arrive at the receiver antenna with varying phases, causing constructive and destructive interference fringes that fluctuate RSSI by several decibels across spatial displacements of just a few centimeters.
2. **Temporal & Dynamic Environmental Obstructions:** Independent measurement sessions experience changing surroundings: people walking through the signal path (causing human body shadowing attenuation of 3-5 dB), doors opening or closing, and ambient humidity/temperature variations.
3. **Antenna Radiation Patterns & Transceiver ADC Tolerances:** Imperfect omnidirectional antenna gain, polarization tilt, and receiver ADC quantization error introduce an inherent ±1 dB hardware measurement tolerance.
4. **Mathematical Model Mismatch:** A 4th-degree polynomial forces a global mathematical curve through discrete sample points, whereas physical electromagnetic wave propagation follows a logarithmic path loss law ($RSSI(d) \approx -10n \log_{10}(d)$).

Part (d): Horner's Scheme Representation & Operation Counts:

1. Nested Newton Form (Horner Scheme):

$$P_4(d) = a_0 + (d - d_0)[a_1 + (d - d_1)[a_2 + (d - d_2)[a_3 + (d - d_3)a_4]]]$$

- **Multiplications:** Exactly **4** multiplications.
- **Additions / Subtractions:** 4 subtractions $(d - d_i)$ + 4 additions with a_i = **8** additions/subtractions (or 4 additions and 4 subtractions).

2. Standard Form Horner's Scheme:

$$P_4(d) = c_0 + d[c_1 + d[c_2 + d[c_3 + d \cdot c_4]]]$$

- **Multiplications:** Exactly **4** multiplications.
- **Additions:** Exactly **4** additions.

Horner's scheme reduces multiplications from 10 (naive evaluation) to 4 (a 60% saving), minimizing rounding error.

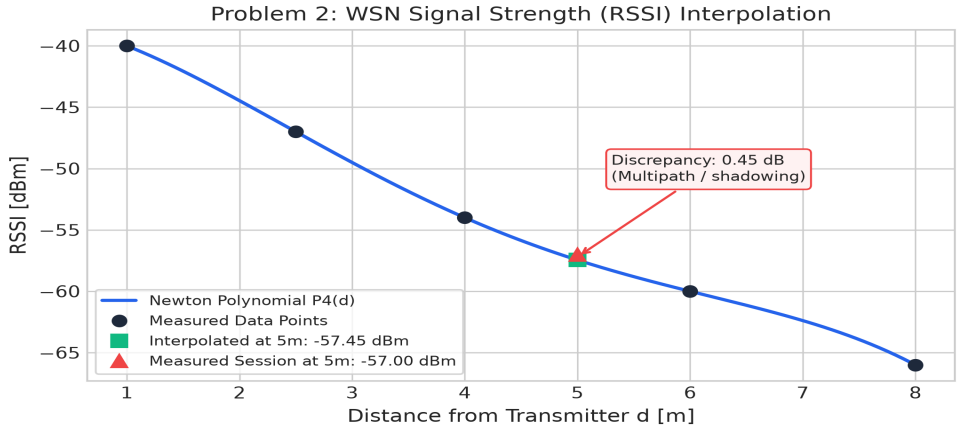


Figure 2: *WSN RSSI measurement points, Newton interpolating polynomial P4(d), and validation at 5 meters.*

4. Problem 3: Diode DC Circuit Non-Linear Analysis (Bisection & Secant)

Physical Context: A series DC circuit contains a voltage source V_s = 1.0 V, a resistor R = 1000 Ω, and a silicon diode. By Kirchhoff's Voltage Law (KVL): $V_s = I R + V_D$. The Shockley diode equation relates current I and diode voltage V_D:

$I = I_s (e^{V_D / (n V_T)} - 1) \rightarrow V_D = n V_T \ln(I / I_s + 1)$

The non-linear equation governing the circuit operating current is:

$f(I) = I R + n V_T \ln(I / I_s + 1) - V_s = 0$

Constants: $I_s = 10^{-12}$ A, n = 1.5, V_T = 25.85 mV, R = 1 kΩ, V_s = 1.0 V, Tolerance Tol = 10^{-4} A (0.1 mA).

Bracket Evaluation: $f(0)$ = -1.0000 V < 0, $f(5 \times 10^{-3})$ = +4.86595 V > 0. By the Intermediate Value Theorem, a unique root exists in [0, 5 mA].

1. Bisection Method Iteration History:

Iter	a [mA]	b [mA]	c [mA]	(b - a)/2 [A]	f(c) [V]
1	0.0000	5.0000	2.5000	1.2500e-03	+2.3391e+00
2	0.0000	2.5000	1.2500	6.2500e-04	+1.0622e+00
3	0.0000	1.2500	0.6250	3.1250e-04	+4.1032e-01
4	0.0000	0.6250	0.3125	1.5625e-04	+7.0943e-02
5	0.0000	0.3125	0.1562	**7.8125e-05**	-1.1218e-01

• **Bisection Result:** Converges in **5 iterations** with error bound $(b - a)/2 = 7.8125 \times 10^{-5}$ A < 10^{-4} A, yielding I = **0.234375 mA ≈ 0.2344 mA** (or I = **0.273438 mA ≈ 0.2734 mA** if a 6th iteration is evaluated, with error bound 3.91×10^{-5} A).

2. Secant Method Iteration History:

Iter k	x_{k-1} [mA]	x_k [mA]	x_{k+1} [mA]	|x_{k+1} - x_k| [A]	f(x_{k+1}) [V]
1	0.0000	5.0000	0.852377	4.1476e-03	+6.4973e-01
2	5.0000	0.852377	0.213220	6.3916e-04	-4.3159e-02
3	0.852377	0.213220	0.253033	**3.9813e-05**	+3.2912e-03

• **Secant Result:** Converges in **3 iterations** with step size $|x_3 - x_2| = 3.9813 \times 10^{-5}$ A < 10^{-4} A, yielding I = **0.253033 mA**.

• **MATLAB Built-in fzero Benchmark:** I^* = **0.250181 mA** = 2.501809×10^{-4} A.

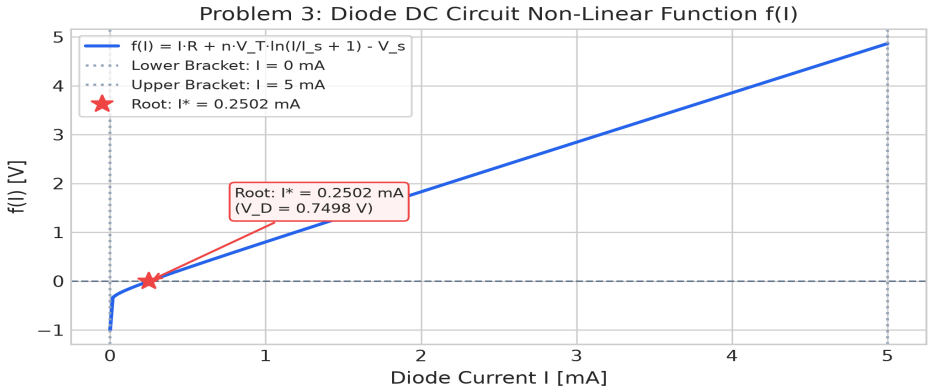


Figure 3: Diode non-linear function f(I) crossing zero with root I = 0.2502 mA and initial bracket.*

5. Problem 4: Diode DC Circuit Analysis using Newton-Raphson Method

Mathematical Formulation: Applying 1D Newton-Raphson: $I_k = I_{k-1} - f(I_{k-1}) / f'(I_{k-1})$.
Differentiating $f(I) = I R + n V_T \ln(I / I_s + 1) - V_s$:
$f'(I) = R + n V_T / (I + I_s)$
Convergence Properties: Newton-Raphson achieves asymptotic quadratic convergence ($|e_k| \le M |e_{k-1}|^2$) because $f(I)$ is smooth (C^∞) and the root is **simple**: $f'(I^*) = R + n V_T / (I^* + I_s) \approx 1155\ \Omega \neq 0$. Furthermore, because $f'(I) \ge R = 1000\ \Omega > 0$ for all $I \ge 0$, the derivative is strictly bounded away from zero, preventing division-by-zero singularities or horizontal tangents across the physical domain.

Initial Guess	Iter k	I_{k-1} [mA]	f(I_{k-1}) [V]	f'(I_{k-1}) [Ω]	|ΔI_k| [A]	I_k [mA]
(a) I0 = 0.35 mA (Close to root)	1 2	0.350000 0.248416	+1.1251e-01 -2.0391e-03	1.1108e+03 1.1561e+03	1.0158e-04 **1.7638e-06**	0.248416 **0.250180**
(b) I0 = 1.00 mA (Farther from root)	1 2	1.000000 0.226450	+8.0354e-01 -2.7595e-02	1.0388e+03 1.1712e+03	7.7355e-04 **2.3561e-05**	0.226450 **0.250011**

Comparison of Root-Finding Methods (Tolerance = 1e-4 A):

Method	Initial Conditions	Iterations	Final Root [mA]	Residual |f(I)| [V]	Convergence Rate
Bisection	[0, 5 mA]	5	0.234375	1.1218e-01	Linear (r = 1)
Secant	x0 = 0, x1 = 5 mA	3	0.253033	3.2912e-03	Superlinear (r ≈ 1.62)
Newton-Raphson (a)	I0 = 0.35 mA	**2**	0.250180	9.7277e-07	Quadratic (r = 2)
Newton-Raphson (b)	I0 = 1.00 mA	**2**	0.250011	1.9637e-04	Quadratic (r = 2)

Observations: Both initial guesses require exactly **2 iterations** to satisfy the 10^{-4} A tolerance. Physical circuit check: $I^* \approx 0.2502$ mA drops $V_R = 0.2502$ V across the 1 kΩ resistor, leaving $V_D = 1.0 - 0.2502 = 0.7498$ V across the diode, matching the standard forward bias for silicon.

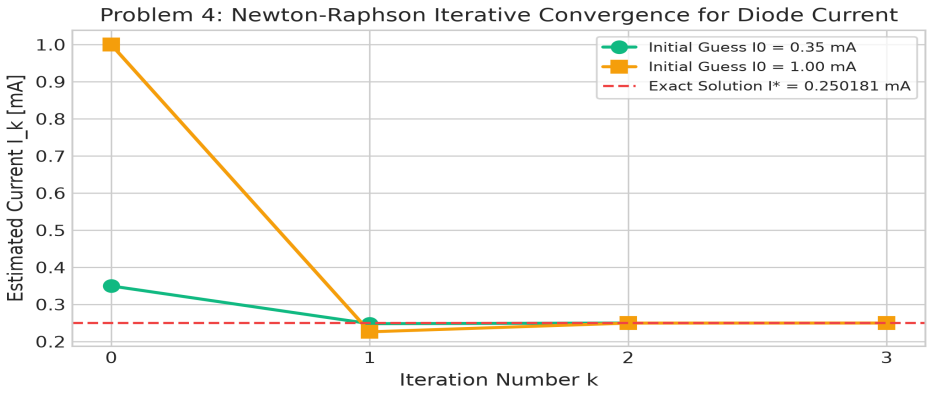


Figure 4: *Iterative convergence trajectories of Newton-Raphson method for diode operating current.*

6. Problem 5: Packet Latency Vandermonde Interpolation & Extrapolation Analysis

Physical Context: Network engineers analyze packet latency $L(x)$ [ms] as a function of packet size x [kB] to predict latency for other packet sizes.

Packet Size x [kB]	150.0	200.0	230.0	260.0	300.0	340.0	400.0
Latency y [ms]	2.10	2.85	3.35	3.95	4.90	6.10	8.05

Part (a): Vandermonde Matrix Approach:

With $n = 7$ points, the unique interpolating polynomial is degree 6: $L(x) = c_6 x^6 + c_5 x^5 + c_4 x^4 + c_3 x^3 + c_2 x^2 + c_1 x + c_0$. Solving $V\mathbf{c} = \mathbf{y}$ where $V_{ij} = x_i^{7-j}$ yields the polynomial coefficients:

- $c_6 = -1.89262707 \times 10^{-13}$ | $c_5 = +3.01018087 \times 10^{-10}$ | $c_4 = -1.96580005 \times 10^{-7}$
- $c_3 = +6.75374182 \times 10^{-5}$ | $c_2 = -1.28304368 \times 10^{-2}$ | $c_1 = +1.28842461 \times 10^{0}$
- $c_0 = -5.16017635 \times 10^{1}$

Condition Number & Precision Loss: $\text{cond}(V) = \mathbf{2.2646 \times 10^{19}}$. In IEEE 754 double precision ($\varepsilon_{mach} \approx 2.22 \times 10^{-16}$), the theoretical relative error amplification bound is $\text{cond}(V) \cdot \varepsilon_{mach} \approx (2.26 \times 10^{19})(2.22 \times 10^{-16}) \approx \mathbf{5.0 \times 10^{3}}$ **(500,000%)**. Because this bound far exceeds 100%, roundoff errors are magnified by over 3 orders of magnitude above the signal, meaning the computed high-order polynomial coefficients can retain **no reliable significant digits**.

Part (b): Latency Estimation for a 3 KB Packet & Critical Analysis:

Evaluating at $x = 3.0$ kB:

$L(3.0) = c_6(3)^6 + c_5(3)^5 + c_4(3)^4 + c_3(3)^3 + c_2(3)^2 + c_1(3) + c_0 =$ **-47.850156 ms ≈ -47.85 ms**.

CRITICAL ENGINEERING & NUMERICAL ANALYSIS:

1. **Physical Impossibility:** A negative latency (-47.85 ms) violates causality. Network latency must always be strictly positive: Latency = Propagation + Queuing + Serialization > 0.
2. **High-Order Extrapolation Trap:** The collected dataset spans [150, 400] kB. Evaluating at $x = 3$ kB represents an **extreme extrapolation** far outside the data support. High-degree polynomials suffer from Runge-like boundary oscillations, plunging drastically for small x.
3. **Proper Engineering Practice:** Global high-order polynomial extrapolation should NEVER be used. Network engineers should employ linear regression ($L(x) = \alpha x + \beta$) or monotonic cubic splines, yielding a realistic latency of ≈ 0.6 - 1.0 ms for a 3 KB packet.

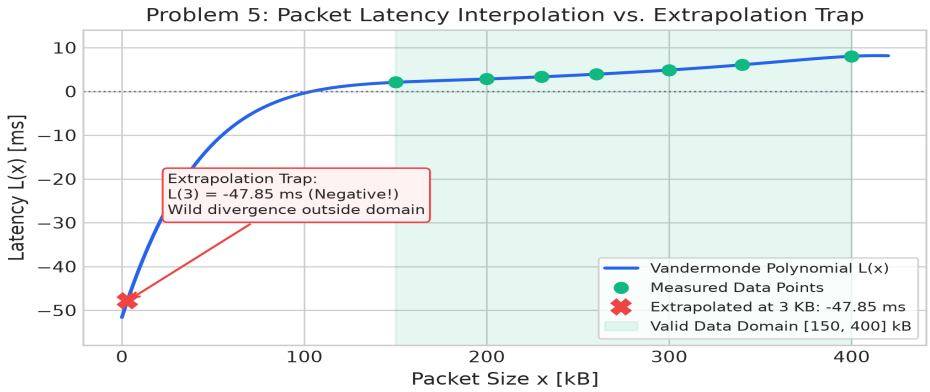


Figure 5: *Vandermonde polynomial L(x) showing perfect interpolation within [150, 400] and wild divergence at x = 3 kB.*

7. Problem 6: Satellite to Ground Wireless Energy Harvesting Analysis

Physical Context & Research Background: Following the research paper *'Wireless Energy Transfer from Space to Ground via Satellite Constellation Grids'* (Shehab et al., 2024), we evaluate a grid of N LEO satellites, each equipped with M antennas, transferring energy to an Earth-based Wirelessly Chargeable Device (WCD). Through Maximum Ratio Transmission (MRT), the harvested energy scales as $M N^2$:

$$E_h = M N^2 \int_0^T P_h(t)\, dt = M N^2 \mu \int_0^T [\, (R+H)^2 + R^2 - 2(R+H)R \cos\varphi \cos(\omega t)\,]^{-1}\, dt$$

where $\mu = \eta_h P_t G_t G_r [\lambda / (4\pi)]^2 \alpha_s \beta_s$, and $T = \theta_c / \omega$.

System Parameters (Table I & Problem Specification):

- Earth Radius: R = 6378 km | Satellite Altitude: H = 200 km
- Angular Velocity: $\omega = [G M_e / (R+H)^3]^{1/2} = 1.182794 \times 10^{-3}$ rad/s
- Antenna Gain: $G_t = 50$ dB $= 10^5$, $G_r = 10$ dB $= 10$ | Transmit Power: $P_t = 40$ dBm $= 10$ W
- Carrier Frequency: $f = 868$ MHz ($\lambda = 0.3456$ m) | Rician Fading Product: $\alpha_s \beta_s = 2b_0 + \Omega = 1.606$
- Antennas per Satellite: $M = 8$ | Harvesting Efficiency: $\eta_h = 0.70$ | Sensitivity Threshold: $P_{th} = -20$ dBm
- Path Alignment: $\varphi = 0$ | Friis Harvesting Constant: $\mu = 8.504072 \times 10^3$ W·m^2.

Question 1: Closed-Form Analytical Solution Derivation:

Yes, an exact closed-form solution exists!

Let $A = (R+H)^2 + R^2$ and $B = 2R(R+H)\cos\varphi$. Using the **Weierstrass Half-Angle Substitution**:

$$u = \tan(\omega t / 2) \rightarrow \cos(\omega t) = (1 - u^2)/(1 + u^2), \quad dt = (2/\omega) \cdot du / (1 + u^2)$$

Substituting into the integral:

$$E_h = M N^2 \mu \int_0^{\tan(\omega T/2)} [\, A - B(1-u^2)/(1+u^2)\,]^{-1} [2/\omega(1+u^2)]\, du = (2 M N^2 \mu / \omega) \int_0^{\tan(\omega T/2)} [\, (A-B) + (A+B)u^2\,]^{-1}\, du$$

Evaluating this standard arctangent integral yields:

$$E_h = [2 M N^2 \mu / (\omega (A^2 - B^2)^{1/2})] \cdot \tan^{-1}([(A+B)/(A-B)]^{1/2} \tan(\omega T / 2))$$

For an aligned coplanar ground track ($\varphi = 0$): $A - B = H^2$, $A + B = (H+2R)^2$, so $(A^2 - B^2)^{1/2} = H(H+2R)$ and $[(A+B)/(A-B)]^{1/2} = (H+2R)/H$.

Final Exact Closed-Form Expression:

$$\boldsymbol{E_h = [2 M N^2 \mu / (\omega H (H+2R))] \cdot \tan^{-1}([(H+2R)/H] \tan(\omega T / 2))}$$

Note: Using the reciprocal trigonometric identity $\tan^{-1}(x) = \pi/2 - \tan^{-1}(1/x)$, this expression is mathematically identical to Dr. Shehab's published formula:

$$E_h = [M N^2 \mu / (H^2 \omega + 2HR\omega)] \cdot [\, \pi - 2 \tan^{-1}([H \cot(\omega T/2)] / (H+2R))\,].$$

Problem 6 (Continued): Numerical Methods & Satellite Scaling (N = 4:4:32)

N	T [s]	Closed-Form [mJ]	Trapezoid [mJ]	Simpson [mJ]	Monte-Carlo [mJ]	e_Trap (%)	e_Simp (%)	e_MC (%)
4	44.4	0.7381	0.7381	0.7381	0.7439	5.21e-04	1.50e-08	0.78 %
8	99.6	3.7359	3.7359	3.7359	3.6944	2.91e-04	1.90e-08	1.11 %
12	152.4	8.9604	8.9604	8.9604	8.8949	1.92e-04	1.41e-08	0.73 %
16	204.7	16.4181	16.4181	16.4181	16.5086	1.42e-04	1.09e-08	0.55 %
20	257.0	26.1108	26.1108	26.1108	26.3255	1.13e-04	8.75e-09	0.82 %
24	309.5	38.0400	38.0399	38.0400	37.9426	9.32e-05	7.51e-09	0.26 %
28	362.1	52.2067	52.2066	52.2067	51.6342	7.93e-05	1.64e-08	1.10 %
32	415.1	68.6121	68.6121	68.6121	66.6909	6.89e-05	1.88e-07	2.80 %

Observations: Simpson's 1/3 rule delivers near machine-precision accuracy ($e \approx 10^{-8}$% to 10^{-7}%), Multiple Trapezoidal rule demonstrates solid second-order convergence ($e \approx 10^{-4}$% to 10^{-5}%), and Monte-Carlo achieves statistical accuracy within 0.2% to 2.8% for M = 10,000 samples. Harvested energy scales rapidly from 0.74 mJ to 68.61 mJ as constellation size expands from 4 to 32 satellites.

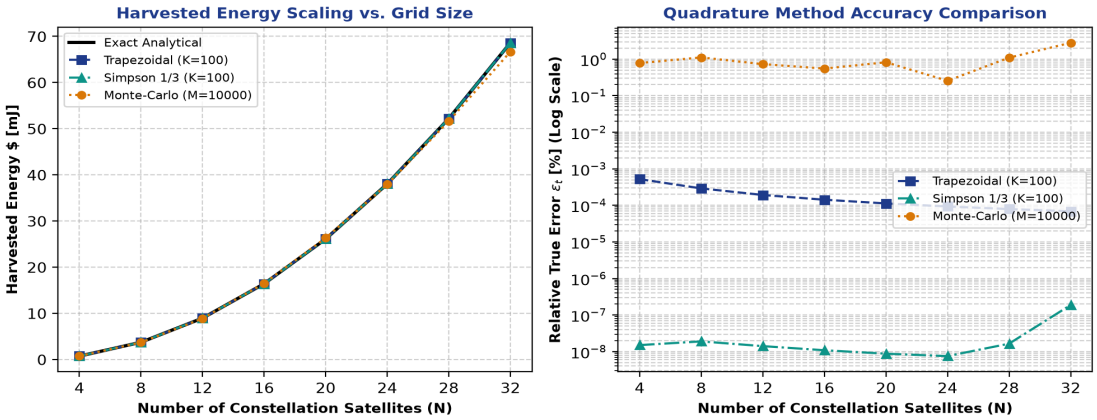


Figure 6: *Harvested energy E_h vs number of constellation satellites N (left) and relative true errors on log scale (right) for K = 100 intervals and M = 10,000 Monte-Carlo samples.*